IGCSE PHYSICS 0625 · TOPIC 1

Motion, Forces & Energy

Measurement, motion, forces, momentum, energy and pressure — key definitions, formulas and graphs, distilled from the syllabus and past papers.

Physical Quantities & Measurement

1.1

DEFINITIONS

Scalar a quantity with magnitude (size) only.

Vector a quantity with magnitude *and* direction.

Scalars	Vectors
distance · speed · time · mass · energy · temperature	force · weight · velocity · acceleration · momentum · electric field strength · gravitational field strength

📌 考过 · ■ 43/M/J/25 Q1 区分标量/矢量 · ■ 41/O/N/22 Q2 弹簧题里的矢量量

Motion

1.2

DEFINITIONS

Speed distance travelled per unit time.

Velocity speed in a given direction.

Acceleration change in velocity per unit time.

FORMULAS

$$v = s \div t$$

speed = distance ÷ time · average speed = total distance ÷ total time

$$a = \Delta v \div \Delta t$$

acceleration = change in velocity ÷ time taken

$$v = u + at \quad \cdot \quad s = ut + \tfrac{1}{2}at^2 \quad \cdot \quad v^2 = u^2 + 2as$$

equations of uniform acceleration

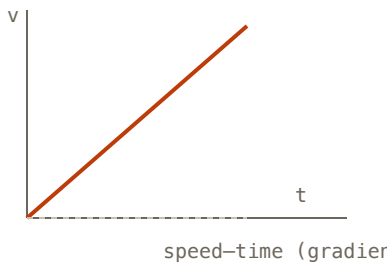
$$g \approx 9.8\ m/s^2$$

acceleration of free fall near the Earth's surface (≈ constant)

GRAPHS



Distance–time — gradient gives speed; a flat line means at rest.



Speed–time — gradient gives acceleration; **area under the line gives distance**.

📌 考过 · ■ 42/F/M/26 Q1 距离–时间图求速度 · ■ 42/F/M/24 Q1 速度–时间图求加速度/距离 · ■ 42/F/M/25 Q4 火车加速度与时间

Mass & Weight

1.3

DEFINITIONS

> **Mass** a measure of the quantity of matter in an object.

> **Weight** a gravitational force on an object that has mass.

> **Gravitational field strength (g)** force per unit mass; equivalent to the acceleration of free fall.

FORMULA

$$W = mg \quad \cdot \quad g = W \div m$$

weight = mass × gravitational field strength (unit: N)

📌 考过 · ■ 42/F/M/26 Q3 北极熊体重 W=mg · ■ 42/M/J/22 Q2 滑轮系统重量与合力

Density

1.4

DEFINITION

> **Density (ρ)** mass per unit volume.

FORMULA

$$\rho = m \div V$$

density = mass ÷ volume (unit: kg/m³ or g/cm³)

📌 考过 · ■ 42/F/M/26 Q2 钢船漂浮——密度 · ■ 42/F/M/22 Q4 密度计算 · ■ 43/M/J/26 Q1 油浮于水

Forces

1.5

DEFINITIONS

Spring constant (k) force per unit extension.

Limit of proportionality the point on a load–extension graph beyond which the graph is no longer a straight line (Hooke's law no longer applies).

Moment force × perpendicular distance from the pivot — a measure of the turning effect of a force.

Centre of gravity the point where the weight of an object appears to act.

Equilibrium no resultant force *and* no resultant moment acting on an object.

FORMULAS

$$F = ma$$

resultant force = mass × acceleration (same direction)

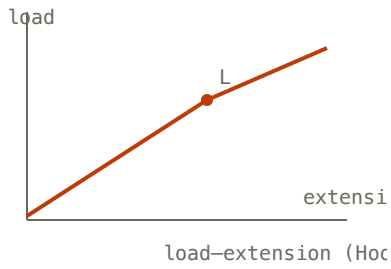
$$k = F \div x$$

spring constant = force ÷ extension (Hooke's law)

$$\text{moment} = F \times d$$

force × perpendicular distance from the pivot (principle of moments: clockwise = anticlockwise)

GRAPH



Load–extension — straight line obeys Hooke's law; L marks the limit of proportionality.

📌 考过 · ■ 42/F/M/25 Q1 弹簧常量与比例极限 · ■ 42/M/J/25 Q2 平衡尺——力矩 · ■ 42/F/M/25 Q4 F=ma · ■ 43/O/N/23 Q2 弹簧与圆周运动

Momentum

1.6

DEFINITIONS

Momentum (p) mass × velocity.

Impulse force × time for which the force acts (= change in momentum).

FORMULAS

$$p = mv$$

momentum = mass × velocity

$$\text{impulse} = F\Delta t = \Delta(mv)$$

impulse = change in momentum

$$F = \Delta p \div \Delta t$$

resultant force = change in momentum per unit time

📌 考过 · ■ 42/F/M/24 Q2 冲量与火箭推力 · ■ 42/F/M/25 Q2 小车碰撞——动量守恒

Energy, Work & Power

1.7

DEFINITIONS

Work done energy transferred when a force moves an object (= F × d).

Power work done per unit time (energy transferred per unit time).

Efficiency useful energy output ÷ total energy input (× 100%).

FORMULAS

$$E_k = \frac{1}{2}mv^2$$

kinetic energy

$$\Delta E_p = mg\Delta h$$

change in gravitational potential energy

$$W = Fd = \Delta E$$

work done = force × distance moved (in the direction of the force)

$$P = W \div t = \Delta E \div t$$

power = work done ÷ time

$$\text{efficiency} = \text{useful output} \div \text{total input}$$

(× 100% for a percentage)

📌 考过 · ■ 42/F/M/24 Q1 动能与刹车 · ■ 43/M/J/26 Q3 碰撞动能变化 · ■ 41/O/N/22 Q1 重力势能与冲量

Pressure

1.8

DEFINITION

> **Pressure** force per unit area.

FORMULAS

$$p = F \div A$$

pressure = force ÷ area (unit: Pa = N/m²)

$$\Delta p = \rho g \Delta h$$

change in pressure beneath a liquid surface = density × g × depth

📌 考过 · ■ 42/F/M/26 Q3 北极熊压强 · ■ 42/F/M/24 Q3 路面与轮胎压强 · ■ 41/M/J/22 Q2 水坝深处压强

Formula summary

Quantity	Formula	Notes
Speed	$v = s/t$	average speed = total distance ÷ total time
Acceleration	$a = \Delta v/\Delta t$	also $v = u + at$, $v^2 = u^2 + 2as$
Weight	$W = mg$	g = gravitational field strength (≈ 9.8 N/kg)
Density	$\rho = m/V$	mass per unit volume
Resultant force	$F = ma$	same direction as acceleration
Hooke's law	$F = kx$	k = spring constant
Moment	moment = Fd	perpendicular distance from pivot
Momentum	$p = mv$	impulse = $F\Delta t = \Delta(mv)$
Kinetic energy	$E_k = \frac{1}{2}mv^2$	—
Gravitational PE	$\Delta E_p = mg\Delta h$	—
Work	$W = Fd$	= energy transferred
Power	$P = W/t$	= energy transferred ÷ time
Pressure	$p = F/A$	liquid: $\Delta p = \rho g \Delta h$